National University of Computer and Emerging Sciences

Karachi Campus

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Attempt all the questions. Attempt all questions according to the sequence, any question answered other than the right sequence will NOT be graded!!

CLO # 1: Apply suitable process models and activities for medium size software systems

Question 1: **[10 Marks]**

1.Briefly describe any two software engineering framework and umbrella activities with examples.

Framework activities are applicable to all software projects, regardless of their size or complexity. In addition, the process framework encompasses a set of umbrella activities that are applicable across the entire software process.

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Framework Activities

•Modeling

o Analysis of requirements

o Design

•Construction

o Code generation

o Testing

Umbrella Activities

•Software project management

•Formal technical reviews

•Software quality assurance

•Software configuration management

•Work product preparation and production

•Reusability management

•Measurement

•Risk management

2.Discuss two advantages and two disadvantages of incremental development.

Functional requirements are prioritized (divided into different groups with different priorities). A number of delivery increments are then defined (corresponding to different groups). The increments are developed either sequentially or in a phased parallel manner. For example – requirements Fr11-Fr15 are included in the first increment, requirements Fr6-Fr10 in the second and requirements Fr1-Fr5 in the third increment

3.Describe three ways in which the Spiral process model is superior to the Waterfall model.

Explicit provision for risk analysis and mitigation; explicit provision for replanning of the process, in mid-stream; explicit consideration at all stages of alternative approaches; supports exploratory prototyping.

4.How Construction of a unit test for a story before coding helpful in XP?

XP Recommends the construction of a unit test for a story before coding commences. So implementer can focus on what must be implemented to pass the test.

5. When using an agile development process, what questions are being focused at a Daily Scrum meeting?

The meeting is guided by what each developer did yesterday, what they will do today, and what obstacles are in their way.

CLO # 2: Analyze software requirements and how to produce software design and architecture

Question 2:

A. Design a user interface form for the "Epic Quest" virtual reality game that allows players to customize their character's appearance. The form should include the following fields.
   - Character Profile: (Character Name, Character Description,
   - Physical Appearance: (Gender, Height, Weight, Hair Color (Blonde, Brown, Red, Black), Eye Color (Blue, Green, Brown, Gray))
   - Save and Reset Options: (Save Changes, Reset to Default)

   Select the most appropriate UI control for each of the fields (Character Name, Description, gender, height, weight, hair color, eye color, save changes, rest to default). Make sure that your UI control design should follow at least 2 design guidelines. Also mention those guidelines. **[10 Marks]**

   Most appropriate UI control for each of the fields
   Character Customization Form
   Section 1: Character Profile
   - Character Name: Text Input Field (single-line text box)
   - Character Description: Text Area (multi-line text box)

   Section 2: Physical Appearance
   - Gender: Radio Buttons (Male/Female)
   - Height: Slider (with units of measurement, e.g., cm/in)
   - Weight: Slider (with units of measurement, e.g., kg/lbs)
   - Hair Color: Drop-down Menu (with options: Blonde, Brown, Red, Black)
   - Eye Color: Drop-down Menu (with options: Blue, Green, Brown, Gray)

   Section 3: Save and Reset Options
   - Save Changes: Button
   - Reset to Default: Button

   Visual Design is must

Solution: 1. Visibility of System Status: The form provides clear and immediate feedback to user interactions, such as showing the selected hair color or eye color.

2. Match Between System and the Real World: The form uses familiar concepts and language, such as sliders for height and weight, and drop-down menus for hair and eye color.

3. User Control and Freedom: The form provides users with clear options to save their changes or reset to default, giving them control over their character customization.

4. Consistency and Standards: The form uses consistent layout, typography, and color schemes throughout, making it easy for users to navigate and understand.

B. A company is developing a new e-commerce platform. The initial requirements included features such as user registration, product catalog, shopping cart, and payment processing. However, during the development phase, the marketing team requests a new feature to integrate social media login and sharing capabilities. The development team estimates that this feature will require significant changes to the existing codebase and may impact the project timeline and budget. Mention the process that should be followed to manage the request for integrating social media login and sharing capabilities into the e-commerce platform). How would you assess the impact of the proposed request on the project? Suppose the request is approved, but the development team encounters technical difficulties that make it challenging to implement. How would you manage this situation and what alternatives would you consider? **[5 Marks]**

i. Mention the process that should be followed to manage the request for integrating social media login and sharing capabilities into the e-commerce platform). Change Management Process

ii. How would you assess the impact of the proposed request on the project?

To evaluate the change request's impact, analyze the requirements to identify affected components, estimate the effort required to implement the change, assess potential risks and mitigation strategies, and document the findings. This comprehensive assessment will inform stakeholders about the change's implications on project scope, timeline, and budget, enabling informed decision-making.

iii. Suppose the request is approved, but the development team encounters technical difficulties that make it challenging to implement. How would you manage this situation and what alternatives would you consider?

To address the technical difficulties, assess the root cause of the issue, collaborate with the development team to find solutions, consider alternative approaches or compromises, and communicate with stakeholders about potential impacts, ensuring transparency and minimizing disruptions.

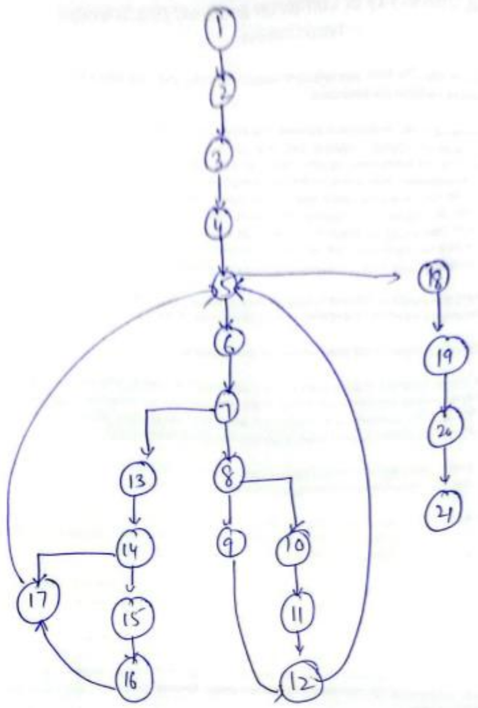
C. A healthcare company is developing a cloud-based patient management system that handles sensitive medical records and real-time patient monitoring data balancing the non- functional requirements of performance and maintainability. Suggest an approach to reconcile the conflicting architectural principles for above mentioned attributes. **[5 Marks]**
Maintainability vs. Performance: Fine-grained components enhance maintainability but might impact performance. Design a modular architecture with coarse-grained components for performance-critical parts and fine-grained components for maintainability-focused areas. Balance component granularity with performance requirements.

CLO # 3: Apply software quality assurance, verification and validation to medium size software systems

Question 3:

A. Construct the control flow graph for the given code. Determine the cyclomatic complexity and identify all the independent paths. Based on these paths, design appropriate test cases. **[10 Marks]**

```
int main() {
    int numbers[] = {4, 60, 9, 11, 52, 3, 8};
    int size = sizeof(numbers) / sizeof(numbers[0]);
    int count = 0;
    for (int i = 0; i < size; i++) {
        int num = numbers[i];
        if (num % 2 == 0) {
            if (num > 50) {
                count += 2;
            } else {
                count += 1;
            }
        } else {
            if (num < 10) {
                count -= 1;
            }
        }
    }
    printf("Final count: %d\n", count);
    return 0;
}
```



cyclomatic = $E - N + 2$
$= 24 - 21 + 2 = 5$ / # of conditions + 1
$4 + 1 = 5$

<u>Paths:</u>

1, 2, 3, 4, 5, 18—21

1— 9, 12, 5, ...

1 —8, 10, 11, 12, 5, ...

1 —7, 13, 14, 15, 16, 17, 5, ...

1 — 7, 13, 14, 17, 5, ...

B. You are part of the quality assurance (QA) team for a tech startup developing **SmartVendo**, an automated vending machine. The machine sells snacks and drinks. The core functionality you're testing is the **purchase flow**, which includes:

- Displaying available items with prices.
- Accepting money (in coins and bills).
- Dispensing selected items.
- Returning change, if necessary.

The development team has just completed the following function: You must define **a minimum of 6 test cases** that thoroughly test the purchase_item function. For each test case, include:

- **Test Case ID**
- **Description**
- **Inputs**: item_code, money_inserted
- **Expected Output**
- **Type of Test** (e.g., Normal, Boundary, Invalid, Edge Case)

The code is given for your reference. **[10 Marks]**

```
def purchase_item(item_code, money_inserted):
    items = {
        "A1": {"name": "Chips", "price": 1.50},
        "B2": {"name": "Soda", "price": 2.00},
        "C3": {"name": "Candy", "price": 1.00}
    }
    if item_code not in items:
        return "Invalid item code"
    price = items[item_code]["price"]
    if money_inserted < price:
        return "Insufficient funds"
    change = round(money_inserted - price, 2)
```

Test Case ID	Description	Inputs	Expected Output	Type of Test
TC01	Valid purchase of Chips with exact amount	`item_code="A1"`, `money_inserted=1.50`	`"Dispensed Chips. Change: $0.0"`	Normal
TC02	Valid purchase of Soda with more money	`item_code="B2"`, `money_inserted=5.00`	`"Dispensed Soda. Change: $3.0"`	Normal
TC03	Invalid item code	`item_code="Z9"`, `money_inserted=2.00`	`"Invalid item code"`	Invalid Input
TC04	Insufficient funds for Candy	`item_code="C3"`, `money_inserted=0.50`	`"Insufficient funds"`	Boundary/Negative
TC05	Purchase Candy with minimal change	`item_code="C3"`, `money_inserted=1.01`	`"Dispensed Candy. Change: $0.01"`	Edge Case
TC06	Case sensitivity in item code (should fail)	`item_code="a1"`, `money_inserted=2.00`	`"Invalid item code"`	Case Sensitivity

C. What are quality attributes? Some external quality attributes are assessed over time, so if I need to do the assessments, what would be the metric for assessment of any four quality attributes? **[5 Marks]**

Property	Measure
Speed	Processed transactions/second User/event response time Screen refresh time
Size	Mbytes ,Number of ROM chips
Ease of use	Training time Number of help frames
Reliability	Mean time to failure Probability of unavailability Rate of failure occurrence Availability
Robustness	Time to restart after failure Percentage of events causing failure Probability of data corruption on failure
Portability	Percentage of target dependent statements Number of target systems

D. Depth of inheritance tree induces complexity and breadth of inheritance tree induces reusability, how? Justify the reasons. **[5 Marks]**

DIT represents the multilevel inheritance which increases the dependence between classes so more coupling, more debugging issues, difficult to understand. Hence complexity is increased whereas, in BIT, the no of siblings' classes increases that represents the code reusability, we can exploit polymorphism, better design can be created

CLO # 4: Understand key principles and common methods for software project management such as scheduling, size estimation, cost estimation and risk analysis

Question 4:

A. The following activities are involved in renovating an office. Each activity has a specific duration and dependencies. Assume project start date is June 1, 2025.

Project Activities

Activity	Task Description	Duration (Days)	Predecessors	Start - End Dates
A	Clear old furniture	4	-	June 1 – June 4

B	Inspect electrical systems	3	-	June 1 – June 3
C	Install new flooring	5	A	June 5 – June 9
D	Paint the walls	6	A	June 5 – June 10
E	Upgrade lighting	4	B	June 4 – June 7
F	Setup new workstations	7	C, E	June 10 – June 16
G	Install glass partitions	5	D	June 11 – June 15
H	Final cleaning and inspection	3	F, G	June 17 – June 19

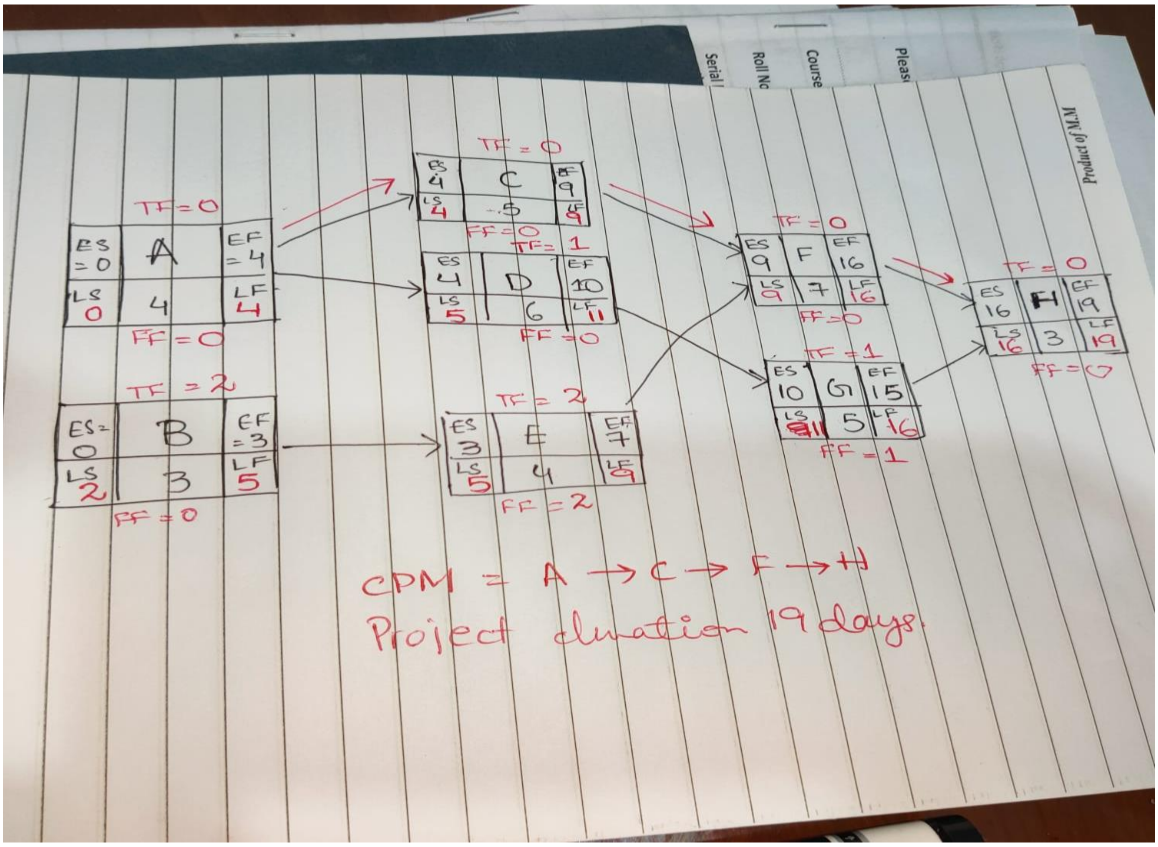
I. Draw the activity network diagram based on the above table. 2

II. Calculate ES, EF, LS, LF, TF and FF for each activity. 6

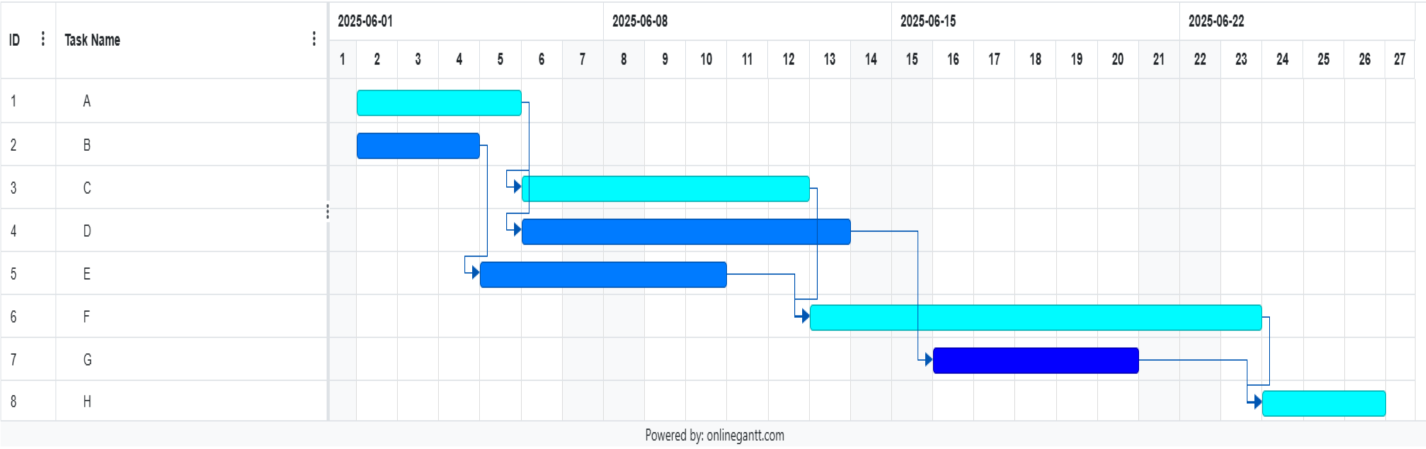
III. Identify the critical path and determine total project duration. 2

IV. Why does a delay in a critical activity affect the entire project duration? 2

V. Given the following activities, their dependencies, Duration and mentioned Start and End dates for a project, design a Gantt chart. 3



1. Because critical activities have zero float, any delay in them directly increases the total project duration as there is no buffer time.



B. You are leading a team working on a Smart City Traffic Monitoring System. The system uses IoT sensors to collect real-time traffic data, sends it to a central server, and processes the data to update digital signboards and a mobile app with live traffic conditions. The system must integrate with multiple external services (e.g., weather updates, public transportation schedules) and comply with government security regulations. The project is on a tight deadline and involves collaboration between local developers and an outsourced team. Recently, you found out that the cloud service provider may increase its rates unexpectedly. One senior developer plans to take a long leave soon. The government authority hasn't finalized the privacy compliance policies, and the outsourced team is still learning the deployment process. There's also been talk of citywide protests that may delay some hardware installations. From the above scenario:

I. Identify five distinct risks present in the project. For each risk, classify it as either a Project Risk, Technical Risk, or Business Risk. 5

II. Briefly justify your classification. 5

III. The hardware installation caused by citywide protests has a high probability of occurrence, rated at 80%. And the impact of this delay is also considered high, with an impact score of 4 on a 5-point scale. The estimated cost incurred for each day of missed installation is Rs. 4500. Using the formula for risk exposure, calculate the total estimated risk cost. 5

Risk Identified	Category	Justification
1. Cloud service provider may increase pricing unexpectedly	Business Risk/project risk	Affects operational cost and long-term budget.
2. Senior developer taking long leave	Project Risk	Affects team velocity and delivery timeline.
3. Privacy compliance policies not finalized	Technical Risk	Impacts design implementation and rework if laws change.
4. Outsourced team unfamiliar with deployment process	Technical Risk	Delays in deployment/testing can arise due to onboarding or integration errors.
5. Citywide protests may delay hardware installations	Project Risk	May postpone site readiness and incur extra labor/time charges.

Risk: Delay due to citywide protests affecting hardware installation
Probability: 80% Impact Scale: 4

Risk Exposure (RE): RE = Probability × Impact = 0.8 × 4 = 3.2
Estimated Cost per Missed Installation Day: Rs. 4500
Total Estimated Risk Cost: = Risk Exposure × Cost per Impact Unit = 3.2 × 4500 = Rs. 14400

C. **BikeWare** is a single-user software application designed for competitive cyclists to track and analyze their riding data. The system allows riders to input two types of data:

- Ride Details: Date, distance (km), duration (mins), temperature (°C), average speed (km/h), and cadence (RPM).
- Rider Profile: Age, weight (kg), and sex (Male/Female).

Both the ride details and rider profiles are recorded in the system. The system processes and outputs the following:

- Four types of graphs (distance, average speed, duration, and calories burned), each available in three time views (daily, weekly, monthly).
- PDF exports for each graph type, generated with different layout logic.

The rider can enquiry the system about Ride history, rider profile display, and a monthly calories-burned summary. System has no external interfaces (EIFs) with any other application.

I. Classify each component as EI, EO, EQ, ILF or EIF, and justify your choice. [4]

II. We assume that each function type in the application has average complexity and that the overall product has a moderate level of complexity. Based on this, calculate the total Function Points (FP) for the application. Additionally, use a productivity rate of 10 hours per Function Point and a cost of Rs. 3500 per hour to compute the total estimated cost of the project. **[6]**

Solution:

EI= 9 {**Ride Details: Date, distance (km), duration (mins), temperature (°C), average speed (km/h), and cadence (RPM) – totaling 6 fields per ride; Rider Profile: Age, weight (kg), and sex (Male/Female) – totaling 3 fields.}**

EO=16 {Four types of graphs (distance, average speed, duration, and calories burned), each available in three time views (daily, weekly, monthly), resulting in 12 graph outputs; PDF exports for each graph type (4 total), generated with different layout logic.]

External inquiries: 3 {Ride history, rider profile display, and a monthly calories-burned summary.}

ILF: 2 {Data is stored in two internal logical files: Ride Records, Rider Profile}

EIF=0 No external interfaces (EIFs) are used.

Measurement Parameter	Count		Weighing factor			
			Simple	Average	Complex	
1. Number of external inputs (EI)	—	*	3	4	6 =	—
2. Number of external Output (EO)	—	*	4	5	7 =	—
3. Number of external Inquiries (EQ)	—	*	3	4	6 =	—
4. Number of internal Files (ILF)	—	*	7	10	15 =	—
5. Number of external interfaces(EIF)	—	*	5	7	10 =	—
Count-total →						

UFC= [9*4+ 16*5 + 3*4+ 2*10+ 0*7]=148

$$F.P = UFP \ X \ CAF$$

$$CAF = 0.65 + (0.01 \ x \sum Fi)$$ Moderately complex product

Then,

$$\sum Fi = 14 \ x \ 2 = 28$$ 2 - Moderate

$$CAF = 0.65 + (0.01 \ x \ 28)$$

$$CAF = 0.93$$

$$F.P = UFP \ X \ CAF$$

FP= 148*0.93= 137.64
I FP requires 10 hrs
137.64 *10= 1376.4 hrs
Per hr rate is Rs. 5000
Total cost=3500*1376.4= Rs. 4817400